

## MBS INTL, MI, USA

WMO: 726379

Lat: 43.533N

Lon: 84.080W

Elev: 201

StdP: 98.93

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14845

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.9      | -15.5 | -22.1                         | 0.5 | -17.0 | -19.6 | 0.7 | -14.8 | 12.2                  | -4.8 | 11.3 | -3.6 | 3.8                   | 260  | 0.580 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.0                   | 31.9            | 22.8 | 30.2 | 21.8 | 28.7 | 21.2 | 24.5                | 29.6 | 23.4 | 28.2 | 22.5 | 26.9 | 5.5                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.8                            | 18.0 | 27.3 | 22.0 | 17.1 | 26.1 | 21.0 | 16.1 | 25.3 | 74.8          | 29.9 | 70.6 | 28.2 | 67.1 | 26.7 | 29.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
|                   |      |     | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 11.1              | 9.6  | 8.6 | DB                         | -20.8 | 34.6               | 3.6 | 1.7                                                | -23.4 | 35.9       | -25.5 | 36.8       | -27.5 | 37.8       | -30.1 | 39.0 |
|                   |      |     | WB                         | -21.1 | 26.3               | 3.5 | 1.2                                                | -23.7 | 27.2       | -25.7 | 28.0       | -27.7 | 28.7       | -30.3 | 29.6 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.0    | -5.1  | -4.1  | 1.1   | 7.7   | 14.5  | 20.1  | 22.0  | 20.9  | 17.2  | 10.6  | 3.9   | -1.8  |  |  |
|                                                                        | DBStd       | 10.74  | 5.97  | 5.79  | 5.94  | 4.90  | 4.82  | 3.93  | 3.13  | 2.99  | 4.08  | 4.78  | 4.92  | 5.19  |  |  |
|                                                                        | HDD10.0     | 1872   | 469   | 396   | 286   | 99    | 11    | 0     | 0     | 0     | 2     | 51    | 192   | 366   |  |  |
|                                                                        | HDD18.3     | 3763   | 727   | 629   | 536   | 319   | 141   | 26    | 5     | 11    | 69    | 243   | 433   | 623   |  |  |
|                                                                        | CDD10.0     | 1497   | 0     | 0     | 8     | 32    | 150   | 302   | 372   | 337   | 216   | 71    | 9     | 1     |  |  |
|                                                                        | CDD18.3     | 347    | 0     | 0     | 1     | 1     | 22    | 77    | 119   | 89    | 34    | 4     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2994   | 0     | 0     | 5     | 23    | 216   | 704   | 1047  | 661   | 296   | 43    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 892    | 0     | 0     | 1     | 3     | 59    | 237   | 337   | 170   | 76    | 8     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.2    | 4.8   | 4.7   | 4.7   | 4.9   | 4.4   | 3.8   | 3.4   | 3.2   | 3.4   | 4.2   | 4.5   | 4.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 789    | 44    | 39    | 60    | 73    | 72    | 70    | 60    | 83    | 99    | 61    | 66    | 57    |  |  |
|                                                                        | PrecMax     | 1012   | 103   | 155   | 170   | 174   | 157   | 131   | 135   | 229   | 410   | 142   | 148   | 135   |  |  |
|                                                                        | PrecMin     | 542    | 14    | 5     | 8     | 29    | 20    | 16    | 11    | 16    | 0     | 12    | 13    | 14    |  |  |
|                                                                        | PrecStd     | 115    | 25    | 31    | 33    | 33    | 34    | 28    | 32    | 47    | 72    | 30    | 32    | 31    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 11.3  | 13.4  | 22.6  | 26.6  | 30.9  | 33.7  | 34.0  | 32.6  | 31.6  | 27.6  | 19.6  | 14.4  |  |  |
|                                                                        |             | MCWB   | 10.1  | 10.4  | 15.7  | 17.2  | 20.3  | 23.4  | 24.4  | 23.6  | 21.8  | 19.5  | 14.0  | 12.0  |  |  |
|                                                                        | 2%          | DB     | 7.6   | 8.7   | 17.3  | 22.5  | 28.4  | 31.4  | 31.9  | 30.2  | 28.8  | 23.9  | 16.2  | 10.0  |  |  |
|                                                                        |             | MCWB   | 6.0   | 6.4   | 12.7  | 14.5  | 19.5  | 21.8  | 23.6  | 22.5  | 20.7  | 17.5  | 12.3  | 8.1   |  |  |
|                                                                        | 5%          | DB     | 4.5   | 5.6   | 13.0  | 19.5  | 25.8  | 29.6  | 30.1  | 28.5  | 26.9  | 21.1  | 13.8  | 7.3   |  |  |
|                                                                        |             | MCWB   | 3.1   | 3.5   | 9.4   | 12.5  | 17.8  | 21.1  | 22.3  | 21.7  | 19.9  | 16.2  | 10.7  | 5.5   |  |  |
|                                                                        | 10%         | DB     | 2.5   | 3.3   | 9.5   | 16.6  | 23.0  | 27.4  | 28.4  | 27.1  | 24.7  | 18.6  | 11.3  | 4.8   |  |  |
|                                                                        |             | MCWB   | 1.4   | 1.5   | 6.3   | 11.1  | 16.6  | 20.1  | 21.3  | 20.7  | 18.7  | 14.6  | 8.7   | 3.1   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 10.4  | 10.8  | 16.5  | 18.9  | 22.5  | 24.7  | 26.3  | 25.4  | 23.9  | 21.3  | 15.3  | 12.6  |  |  |
|                                                                        |             | MCDB   | 11.1  | 12.3  | 22.6  | 24.6  | 28.4  | 31.4  | 31.9  | 30.6  | 29.1  | 25.6  | 18.1  | 14.4  |  |  |
|                                                                        | 2%          | WB     | 6.1   | 6.6   | 13.0  | 15.8  | 20.6  | 23.3  | 24.9  | 23.9  | 22.3  | 19.0  | 13.1  | 8.6   |  |  |
|                                                                        |             | MCDB   | 7.4   | 8.7   | 16.4  | 20.3  | 25.6  | 29.4  | 29.9  | 28.1  | 26.3  | 22.1  | 15.7  | 9.8   |  |  |
|                                                                        | 5%          | WB     | 3.3   | 3.6   | 9.7   | 13.7  | 19.2  | 22.3  | 23.7  | 22.9  | 21.0  | 16.8  | 11.0  | 5.7   |  |  |
|                                                                        |             | MCDB   | 4.2   | 5.4   | 12.6  | 18.4  | 24.1  | 27.6  | 28.4  | 26.9  | 24.6  | 20.5  | 13.5  | 7.2   |  |  |
|                                                                        | 10%         | WB     | 1.5   | 1.7   | 6.4   | 11.5  | 17.6  | 21.2  | 22.6  | 21.9  | 19.8  | 15.0  | 8.7   | 3.2   |  |  |
|                                                                        |             | MCDB   | 2.4   | 3.2   | 9.4   | 16.2  | 22.2  | 25.6  | 26.7  | 25.5  | 23.5  | 18.2  | 11.0  | 4.6   |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 6.9   | 7.7   | 8.8   | 10.6  | 10.9  | 10.9  | 11.0  | 10.7  | 11.5  | 9.9   | 7.6   | 6.2   |  |  |
|                                                                        | 5% DB       | MCDBR  | 8.4   | 9.2   | 13.6  | 15.4  | 13.7  | 12.9  | 12.5  | 12.1  | 13.9  | 13.7  | 10.7  | 8.6   |  |  |
|                                                                        |             | MCWBR  | 7.6   | 7.5   | 9.4   | 9.2   | 7.1   | 5.8   | 6.0   | 5.9   | 6.8   | 8.0   | 8.1   | 7.4   |  |  |
|                                                                        | 5% WB       | MCDBR  | 7.8   | 8.9   | 12.9  | 13.8  | 11.9  | 11.1  | 10.7  | 10.1  | 11.1  | 12.3  | 10.0  | 8.4   |  |  |
|                                                                        |             | MCWBR  | 7.2   | 7.7   | 9.5   | 9.6   | 7.2   | 6.0   | 5.9   | 5.6   | 6.4   | 8.0   | 8.3   | 7.7   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.291 | 0.317 | 0.331 | 0.365 | 0.390 | 0.419 | 0.407 | 0.400 | 0.374 | 0.341 | 0.311 | 0.284 |  |  |
|                                                                        | taud        |        | 2.379 | 2.304 | 2.386 | 2.317 | 2.302 | 2.252 | 2.292 | 2.306 | 2.377 | 2.462 | 2.492 | 2.446 |  |  |
|                                                                        | Ebn at Noon |        | 840   | 873   | 909   | 899   | 882   | 854   | 860   | 854   | 851   | 839   | 815   | 818   |  |  |
|                                                                        | Edh at Noon |        | 78    | 100   | 105   | 122   | 127   | 135   | 128   | 122   | 105   | 85    | 70    | 67    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.40  | 2.29  | 3.58  | 4.57  | 5.38  | 6.04  | 6.09  | 5.18  | 4.14  | 2.59  | 1.57  | 1.11  |  |  |
|                                                                        | RadStd      |        | 0.14  | 0.28  | 0.38  | 0.39  | 0.48  | 0.34  | 0.30  | 0.23  | 0.40  | 0.26  | 0.20  | 0.10  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +108    | N/A     |
| (47) Regional (58 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page